

NanoGT Match Report: Dravet syndrome

Disease: Dravet syndrome (ORPHA:1306)

Primary gene: SCN1A

Gene CDS: 6027 bp

Inheritance: Autosomal dominant

Target tissues scored: CNS

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: MULTI-SUBUNIT ENZYME (SCN1A): the disease gene encodes one subunit of a multi-polypeptide enzyme complex. Scoring addresses this subunit only. Confirm whether restoring this single subunit reconstitutes full enzymatic activity, or whether co-delivery of other complex subunits is required.; Vector does not naturally cover all annotated disease tissues: cns; Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy.

Disease Mechanism Evidence

Molecular mechanism: haploinsufficiency

Mechanistic detail: Haploinsufficiency of Nav1.1 sodium channel (SCN1A) preferentially reduces firing in fast-spiking GABAergic inhibitory interneurons causing disinhibition and refractory seizures

Gene-addition compatibility: conditional

Preferred modality class: lv or aav inhibitory interneuron targeted scn1a

Evidence level/status: direct / source_checked

Evidence summary: Claes L et al. (2003 Hum Mutat PMID 12754708) established de novo SCN1A haploinsufficiency as the primary mechanism; Colasante G et al. (2020 Mol Ther PMID 31607539) demonstrated that cell-type-specific SCN1A upregulation restores interneuron excitability; gene addition is conditional because ectopic expression in excitatory neurons would worsen seizures

Evidence source: Claes L et al. 2003 Hum Mutat PMID 12754708; Colasante G et al. 2020 Mol Ther PMID 31607539

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
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Top 4 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Skysona	LV	7.1/10	Medium	approved

Rank	Program	Vector	Score	Confidence	Approval
2	Libmeldy	LV	6.6/10	Medium	approved
3	Strimvelis	LV	6.0/10	Medium	approved
4	AVR-RD-01	LV	5.8/10	Medium	phase1/2

Match #1: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 7.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	7.10	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 6027bp / cargo 8000bp (75% utilized)
- Precedent target match: cns

- Both membrane proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Haploinsufficiency of Nav1.1 sodium channel (SCN1A) preferentially reduces firing in fast-spiking GABAergic inhibitory interneurons causing disinhibition and refractory seizures
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Claes L et al. (2003 Hum Mutat PMID 12754708) established de novo SCN1A haploinsufficiency as the primary mechanism; Colasante G et al. (2020 Mol Ther PMID 31607539) demonstrated that cell-type-specific SCN1A upregulation restores interneuron excitability; gene addition is conditional because ectopic expression in excitatory neurons would worsen seizures
- Mechanism source: Claes L et al. 2003 Hum Mutat PMID 12754708; Colasante G et al. 2020 Mol Ther PMID 31607539 (<https://pubmed.ncbi.nlm.nih.gov/12754708/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — SCN1A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- MULTI-SUBUNIT ENZYME (SCN1A): the disease gene encodes one subunit of a multi-polypeptide enzyme complex. Scoring addresses this subunit only. Confirm whether restoring this single subunit reconstitutes full enzymatic activity, or whether co-delivery of other complex subunits is required.
- Vector does not naturally cover all annotated disease tissues: cns
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #2: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.6 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.62	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 6027bp / cargo 8000bp (75% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Haploinsufficiency of Nav1.1 sodium channel (SCN1A) preferentially reduces firing in fast-spiking GABAergic inhibitory interneurons causing disinhibition and refractory seizures
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Claes L et al. (2003 Hum Mutat PMID 12754708) established de novo SCN1A haploinsufficiency as the primary mechanism; Colasante G et al. (2020 Mol Ther PMID 31607539) demonstrated that cell-type-specific SCN1A upregulation restores interneuron excitability; gene addition is conditional because ectopic expression in excitatory neurons would worsen seizures

- Mechanism source: Claes L et al. 2003 Hum Mutat PMID 12754708; Colasante G et al. 2020 Mol Ther PMID 31607539 (<https://pubmed.ncbi.nlm.nih.gov/12754708/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — SCN1A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- MULTI-SUBUNIT ENZYME (SCN1A): the disease gene encodes one subunit of a multi-polypeptide enzyme complex. Scoring addresses this subunit only. Confirm whether restoring this single subunit reconstitutes full enzymatic activity, or whether co-delivery of other complex subunits is required.
- Vector does not naturally cover all annotated disease tissues: cns
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #3: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 6.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class

Dimension	Score	Max	What it measures
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.05	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 6027bp / cargo 8000bp (75% utilized)
- No tissue overlap (disease: ['CNS'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Haploinsufficiency of Nav1.1 sodium channel (SCN1A) preferentially reduces firing in fast-spiking GABAergic inhibitory interneurons causing disinhibition and refractory seizures
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Claes L et al. (2003 Hum Mutat PMID 12754708) established de novo SCN1A haploinsufficiency as the primary mechanism; Colasante G et al. (2020 Mol Ther PMID 31607539) demonstrated that cell-type-specific SCN1A upregulation restores interneuron excitability; gene addition is conditional because ectopic expression in excitatory neurons would worsen seizures
- Mechanism source: Claes L et al. 2003 Hum Mutat PMID 12754708; Colasante G et al. 2020 Mol Ther PMID 31607539 (<https://pubmed.ncbi.nlm.nih.gov/12754708/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expres-

sion expected

- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — SCN1A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- MULTI-SUBUNIT ENZYME (SCN1A): the disease gene encodes one subunit of a multi-polypeptide enzyme complex. Scoring addresses this subunit only. Confirm whether restoring this single subunit reconstitutes full enzymatic activity, or whether co-delivery of other complex subunits is required.
- Vector does not naturally cover all annotated disease tissues: cns
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #4: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 5.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector

Dimension	Score	Max	What it measures
Therapeutic window	1.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.81	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 6027bp / cargo 8000bp (75% utilized)
- No tissue overlap (disease: ['CNS'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Haploinsufficiency of Nav1.1 sodium channel (SCN1A) preferentially reduces firing in fast-spiking GABAergic inhibitory interneurons causing disinhibition and refractory seizures
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Claes L et al. (2003 Hum Mutat PMID 12754708) established de novo SCN1A haploinsufficiency as the primary mechanism; Colasante G et al. (2020 Mol Ther PMID 31607539) demonstrated that cell-type-specific SCN1A upregulation restores interneuron excitability; gene addition is conditional because ectopic expression in excitatory neurons would worsen seizures
- Mechanism source: Claes L et al. 2003 Hum Mutat PMID 12754708; Colasante G et al. 2020 Mol Ther PMID 31607539 (<https://pubmed.ncbi.nlm.nih.gov/12754708/>)
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — SCN1A standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- MULTI-SUBUNIT ENZYME (SCN1A): the disease gene encodes one subunit of a multi-polypeptide enzyme complex. Scoring addresses this subunit only. Confirm whether restoring this single subunit reconstitutes full enzymatic activity, or whether co-delivery of other complex subunits is required.
- Vector does not naturally cover all annotated disease tissues: cns
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Excluded Programs (Packaging Failure)

Program	Vector	Gene CDS Issue
ABO-101	AAV9	Gene CDS (6027bp) exceeds vector cargo (4700bp) — hard fail
AT132	AAV8	Gene CDS (6027bp) exceeds vector cargo (4700bp) — hard fail
BMN 307	AAV5	Gene CDS (6027bp) exceeds vector cargo (4700bp) — hard fail
CPCB-RPE1	AAV8	Gene CDS (6027bp) exceeds vector cargo (4700bp) — hard fail
DTX201	AAV8	Gene CDS (6027bp) exceeds vector cargo (4700bp) — hard fail